Predicting Myocardial Infarction and Coronary Heart Disease from BRFSS Survey Features

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***Abstract*—We predict myocardial infarction or coronary heart disease (MICHD) from the 2015 Behavioral Risk Factor Surveillance System: 328,135 labeled respondents, 321 survey features, and a positive rate of 8.8%. Because accuracy is dominated by the negative class, model selection uses the F1 score of the disease class (F1+). All estimators are implemented in NumPy from the CS-433 lectures. On a held-out validation fold we compare six linear methods and retain logistic regression trained by batch gradient descent (200 steps, step size 0.1) with decision threshold $t^* = 0.23$ (validation F1+ $= 0.4176$). The same hyperparameters are then used to refit preprocessing and weights on the full training set. The resulting submission attains public F1+ $= 0.432$ (accuracy 0.871) on AIcrowd. The unlabeled test set is used only for this final prediction.**

I. Introduction

We treat MICHD screening as binary classification: $y = +1$ if a BRFSS respondent reports myocardial infarction or coronary heart disease, and $y = -1$ otherwise [1], [2]. Of 328,135 training labels, 28,975 are positive. A constant predictor of -1 therefore reaches accuracy 0.911 with zero recall on the disease class, so we report F1+, the metric used by the course leaderboard.

The implementation comprises the six required estimators in `implementations.py` [3], a leakage-aware preprocessor in `src/preprocess.py`, and a two-phase driver `run.py` controlled by `config.yaml`. Phase 1 selects the method and the decision threshold on a validation split. Phase 2 refits the chosen model on all labeled rows and writes the competition file. This report follows the Project 1 IEEE layout [4].

II. Models and Methods

Data and preprocessing.: The training matrix has 328,135 rows and 321 columns; the test matrix has 109,379 rows. About 44.8% of entries are missing before cleaning. Following an exploratory analysis (`doc/eda.ipynb`), we discard 143 columns and retain 178: survey identifiers and sampling weights, duplicate anthropometric fields, modules with more than 90% missing values, and leakage items such as `HAREHAB1` that are consequences of a cardiac diagnosis rather than predictors. BRFSS sentinel codes (77, 99, 777, and analogues) are mapped to missing. Codes 7 and 9 are treated as "don't know" or "refused" except on age variables, where they are valid groups; for `_AGEG5YR` only code 14 is missing. For `PHYSHLTH`, `MENTHLTH`, `POORHLTH`, and `CHILDREN`, code 88 (none) is mapped to zero. Each retained column is imputed by its training median and standardized to zero mean and unit variance on the same rows. A column of ones is then prepended, so the linear predictor is $x^\top w$ with an intercept.

Evaluation protocol.: Labeled rows are shuffled with seed 0 and split 80/20 into a training fold and a validation fold *before* any median or scale is computed, so validation rows do not enter the preprocessing statistics. Phase 1 trains each candidate on the 80% fold only. For every fitted weight vector we evaluate 51 decision thresholds on an evenly spaced grid $t \in [0.05, 0.55]$ and retain the value t^* that maximizes validation F1+. For ridge regression the same search is repeated over 12 values of the penalty, $\lambda \in [10^{-4}, 10]$, on a logarithmic grid. Phase 2 does not repeat this search. It refits the preprocessor and the selected estimator on all 328,135 labeled rows, using the hyperparameters chosen in phase 1, and applies the fixed threshold to the test scores. Test labels are never observed.

Metrics.: Let TP, FP, TN, and FN be the confusion counts for the disease class $y = +1$, and let $N = TP + TN + FP + FN$. Accuracy, positive-class recall, precision, and F1 are

$$\mathrm{Acc} = \frac{TP + TN}{N}, \quad \mathrm{Rec}^+ = \frac{TP}{TP + FN}, \quad \mathrm{Prec}^+ = \frac{TP}{TP + FP}, \quad \mathrm{F1}^+ = \frac{2\,\mathrm{Prec}^+\,\mathrm{Rec}^+}{\mathrm{Prec}^+ + \mathrm{Rec}^+}. \tag{1}$$

Model selection uses $\mathrm{F1}^+$.

Candidate estimators.: All six candidates are linear in the preprocessed features. The first four use labels in $\{+1, -1\}$ and the mean squared error

$$L(w) = \frac{1}{2N}\|y - Xw\|_2^2, \quad \nabla L(w) = \frac{1}{N} X^\top (Xw - y). \tag{2}$$

Their updates are

$$w \leftarrow w - \gamma \nabla L(w), \tag{3}$$

$$w \leftarrow w - \gamma\, x_n (x_n^\top w - y_n), \tag{4}$$

$$X^\top X\, w = X^\top y, \tag{5}$$

$$(X^\top X + 2N\lambda I)\, w = X^\top y. \tag{6}$$

(3) is batch GD with $\gamma = 0.1$ for 200 steps. (4) is SGD: one row n is drawn at each step, with $\gamma = 0.01$ for 200 steps. (5)

Table I
VALIDATION COMPARISON (PHASE 1, 20% HOLDOUT, SEED 0).

Method	t^*	Val F1+	Setting
Least squares	0.05	0.0624	closed form
MSE, batch GD	0.05	0.0606	200 steps, $\gamma = 0.1$
MSE, SGD	0.05	0.1435	200 steps, $\gamma = 0.01$
Ridge	0.05	0.3782	best $\lambda = 3.51$
Reg. logistic	0.23	0.4179	$\lambda = 2.85 \times 10^{-4}$
Logistic	0.23	**0.4176**	200 steps, $\gamma = 0.1$

is least squares. (6) is ridge regression, which adds $\lambda\|w\|_2^2$ to (2); λ is chosen on the validation grid. These four models predict $+1$ when $x^\top w \geq t$.

Logistic regression maps labels to $\{0, 1\}$ and uses $\sigma(z) = (1+e^{-z})^{-1}$ and $\hat{p}(y = 1 \mid x) = \sigma(x^\top w)$. The mean negative log-likelihood and its gradient are

$$\begin{aligned} L(w) &= -\frac{1}{N}\sum_{n=1}^{N}\Big[y_n \log \sigma(x_n^\top w) \\ &\qquad + (1 - y_n)\log\big(1 - \sigma(x_n^\top w)\big)\Big], \\ \nabla L(w) &= \frac{1}{N} X^\top \big(\sigma(Xw) - y\big). \end{aligned} \tag{7}$$

Weights start at zero and take 200 steps of size $\gamma = 0.1$. Regularized logistic regression uses the same iteration on

$$\nabla L(w) + 2\lambda w, \qquad \lambda = 2.85 \times 10^{-4}, \tag{8}$$

with λ fixed from a prior validation sweep; phase 1 then selects only t^*. Both logistic models predict $+1$ when $\hat{p} \geq t$.

Selected model.: Unregularized logistic regression is the submitted system, with $t^* = 0.23$. Phase 2 repeats that optimization on all labeled rows and applies the same threshold to the test probabilities. The regularized variant is reported as a comparison and is not the uploaded model.

III. RESULTS

Table I reports phase-1 validation F1+ for the six candidates under the protocol above. Squared-error methods remain weak once the threshold is chosen for F1+: least squares and full-batch MSE reach about 0.06, and 200 steps of SGD reach 0.144. Ridge regression improves to 0.378 at $\lambda = 3.51$ and $t^* = 0.05$. Regularized logistic regression, at the fixed penalty $\lambda = 2.85 \times 10^{-4}$ and $t^* = 0.23$, reaches 0.4179. Logistic regression at $t^* = 0.23$ attains validation F1+ $= 0.4176$ (accuracy 0.865, recall 0.543, precision 0.339). The default cutoff 0.5 on the same probabilities yields F1+ $= 0.192$, because most positive-class scores lie below one half.

Table II records public test F1+ for successive logistic submissions. The bold row is the submitted system. Earlier files used fewer iterations, a threshold of 0.5 or 0.36, weights fit on 80% of the training data, or the regularized gradient. The test set was not used to choose among these settings.

Table II
PUBLIC TEST F1+ FOR LOGISTIC SUBMISSIONS.

Setting	F1+
50 steps, $t = 0.5$, fit on 80% of train	0.261
50 steps, $t = 0.36$	0.419
Reg. logistic, $\lambda = 2.85 \times 10^{-4}$, $t^* = 0.23$	0.431
200 steps, $t^* = 0.23$, refit on all train	**0.432**

IV. DISCUSSION

Validation and public test disagree only slightly (0.4176 versus 0.432), which is consistent with refitting on the rows that were held out during tuning. Accuracy on the public test is 0.871 and remains a poor summary of the same predictions. The gain from $t = 0.5$ to $t^* = 0.23$ shows that class imbalance is handled here by the decision threshold, not by changing the loss. Squared-error estimators, including a tuned ridge penalty, do not match the logistic likelihood on this rare positive class. Adding $\lambda = 2.85 \times 10^{-4}$ to the logistic gradient changes validation F1+ only in the fourth decimal (0.4179 versus 0.4176). On the public test the same penalty scores 0.431 (accuracy 0.870), below the unregularized submission (0.432, accuracy 0.871), so it does not replace that file. Further improvement would require representations that are not linear in the raw survey codes; products of standardized codes did not raise validation F1+ in our checks.

V. SUMMARY

We select logistic regression by validation F1+ and submit a full-data refit at $t^* = 0.23$. To reproduce the file, place the three competition CSVs in `data/`, set `config.yaml` to `phase: 2`, `method: logistic`, and `threshold: 0.23`, run `python run.py`, and upload `outputs/submission_baseline.csv`. The corresponding commit URL is submitted with the course form at http://mlcourse.epfl.ch.

REFERENCES

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